30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 22 and 23 (Ch. 17)

The $IS - LM$ with expectations

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
   - Elements of national accounting

2. The short run
   1. Goods markets
   2. Financial markets
      1. Without commercial banks
      2. With commercial banks
   3. IS-LM model

3. The medium run
   1. Labor market
   2. Phillips curve
   3. IS-LM-PC model
      1. Medium and short run equilibrium
      2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
   1. Financial markets and expectations
   2. Consumption, investment, and expectations
   3. Output, economic policies, and expectations

5. Open economy
   1. Exchange rates, trade, and international returns
   2. Goods markets in an open economy
   3. Financial markets and economic policies in an open economy

6. Public debt and fiscal policy

7. From the Great Recession to Quantitative Tightening
   - From a housing crisis to a financial crisis
   - Unconventional monetary policies

TWO PERIODS IS – LM MODEL

Add expectations in the IS –LM model using the intertemporal approach

AIM:

- explain interactions between expectations and real and financial variables

- Show how expectations about the future influence the current dynamics of the economy

Assumptions

- Two periods: present (t) and future (t+1) (all periods after t)

- In both periods all the markets are in equilibrium

  - equilibrium in the goods market: IS curve

$$Y = C + I + G$$

  - Equilibrium in the money market: LM curve with endogenous (standard case) or exogenous (alternative case) monetary policy

$$M^S = M^D$$

- t+1 is the last period => expectations about the future play a role only in current period

- Rational agents: use in efficient way available information t : current period

The goods market: expectations influence only the goods market through C and I

- Aggregate consumption

$$C_t = C(Y_t - T_t, Y_{t+1}^e - T_{t+1}^e, FW_t, HW_t, r_t)$$

- Aggregate investment

$$I_t = f(\Pi_t, \mathrm{V}(\Pi^e), x_t, x_{t+1}^e) = f(\Pi_t, \Pi_{t+1}^e, r_t, r_{t+1}^e, \delta, x_t, x_{t+1}^e)$$

$\Rightarrow$ The goods market is in equilibrium if

$$Y_t = \underbrace{C_t + I_t} + G_t$$

A_t = Aggregate private expenditure:
depends on current and expected future variables

If

- $Y_t \uparrow$
  - $LW_t \uparrow \Rightarrow$ more consumption opportunities over the life cycle $\Rightarrow$ $C_t \uparrow$
  - $\Pi_t \uparrow \Rightarrow$ firms are more willing to invest, as they expect higher returns $\Rightarrow$ $I_t \uparrow$

=> aggregare private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $Y_{t+1}^e \uparrow$
  - more disposable income next period but more consumption opportunities over the life cycle $\Rightarrow LW_t \uparrow \Rightarrow C_t \uparrow$
  - Fims expect higher profits next period and the PDV of the stream of profits $\uparrow \Rightarrow$ firms are more willing to invest, as they expect higher returns $\Rightarrow$ $I_t \uparrow$
  - Stocks become more attractive for investors as they expect higher dividends $€D_{t+1}^e \uparrow \Rightarrow$ higher demand and their price increases $\Rightarrow €Q_t \uparrow \Rightarrow$ $FW_t \uparrow$

=> aggregate private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $x_t \downarrow$ and/or $x_{t+1}^e \downarrow$
  - The borrowing rate today or tomorrow is lower $\Rightarrow$ investment is cheaper for firms $\Rightarrow$ $I_t \uparrow$

$\Rightarrow$aggregate private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $T_t \downarrow$ and/or $T_{t+1}^e \downarrow$
  - More resources acre available over the life cycle $\Rightarrow$ $LW_t \uparrow \Rightarrow$ more consumption opportunities $\Rightarrow$ $C_t \uparrow$

$\Rightarrow$aggregate private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $r_t \downarrow$
  - Given disposable income in both periods its PDV increases $\Rightarrow$more consumption opportunities over the life cycle $\Rightarrow LW_t \uparrow \Rightarrow C_t \uparrow$
  - $\uparrow$ the borrowing rate decreases and the PDV of expected profits increases making investment in physical capital more convenient for firms more convenient to invest $\Rightarrow \ I_t$
  - Stocks become more attractive than bonds for investors $\Rightarrow$ higher demand for stocks and their price increases $\Rightarrow €Q_t \uparrow \Rightarrow \ FW_t \uparrow \Rightarrow \ C_t \uparrow$

=> aggregare private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $r_{t+1}^e \downarrow$
  - Stocks become more attractive than bonds for investors $\Rightarrow$ higher demand for stocks and their price increases $\Rightarrow €Q_t \uparrow \Rightarrow \ FW_t \uparrow$
  - firms are more willing to invest since the PDV of expected profits increases $\Rightarrow \mathrm{V}(\Pi^e) \uparrow \Rightarrow \ I_t \uparrow$

=> aggregare private expenditure inreases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

- $\delta \downarrow$ firms are more willing to invest since the devaluation process is slower and the PDV of expected profits increases $\Rightarrow \mathrm{V}(\Pi^e) \uparrow \Rightarrow \ I_t \uparrow$

$\Rightarrow$aggregate private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

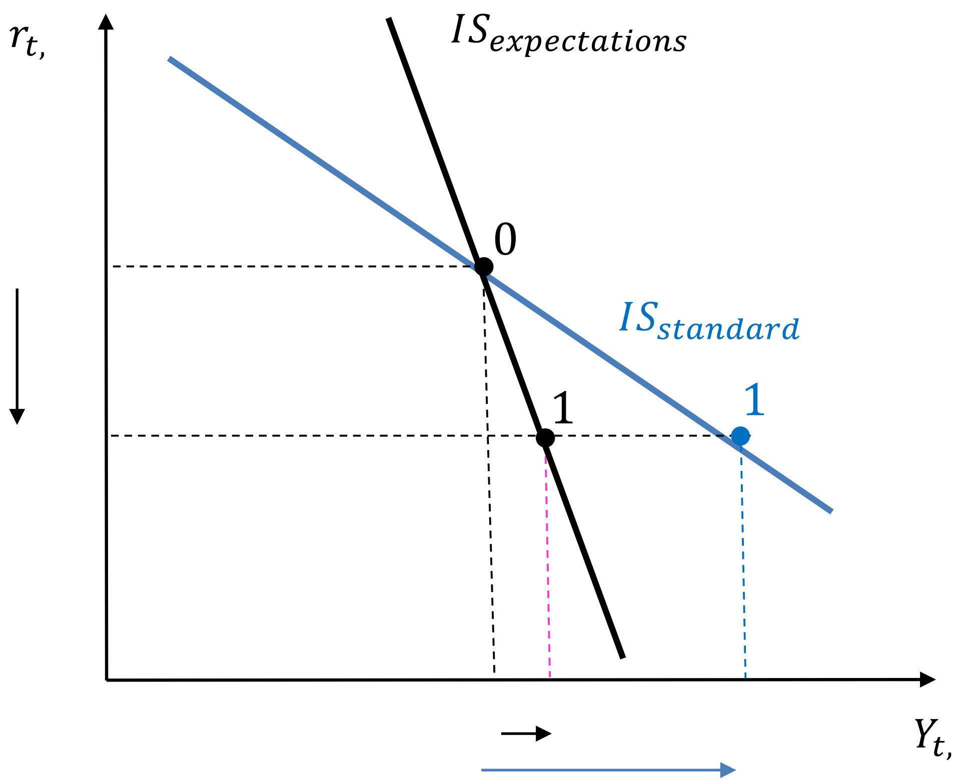
In a synthetic way the current IS With expectations will be

$$Y_t = \underbrace{C\left(Y_{t,}, T_t, Y^e_{t+1}, T^e_{t+1}, r_{t,}\right) + I\left(Y_{t,}, Y^e_{t+1}, r_{t,} r^e_{t+1}, x_{t,} x^e_{t+1}, \delta\right)}_{A_t} + G_t$$

$Y_t \uparrow Y^e_{t+1} \Rightarrow A_t \uparrow$
$r_t \downarrow\ r^e_{t+1} \downarrow \Rightarrow A_t \uparrow\ T_t$
$\downarrow\ T^e_{t+1} \downarrow \Rightarrow A_t \uparrow$

$x \downarrow \Rightarrow A_t \uparrow$
$\delta \downarrow \Rightarrow A_t \uparrow$

Current period (t)

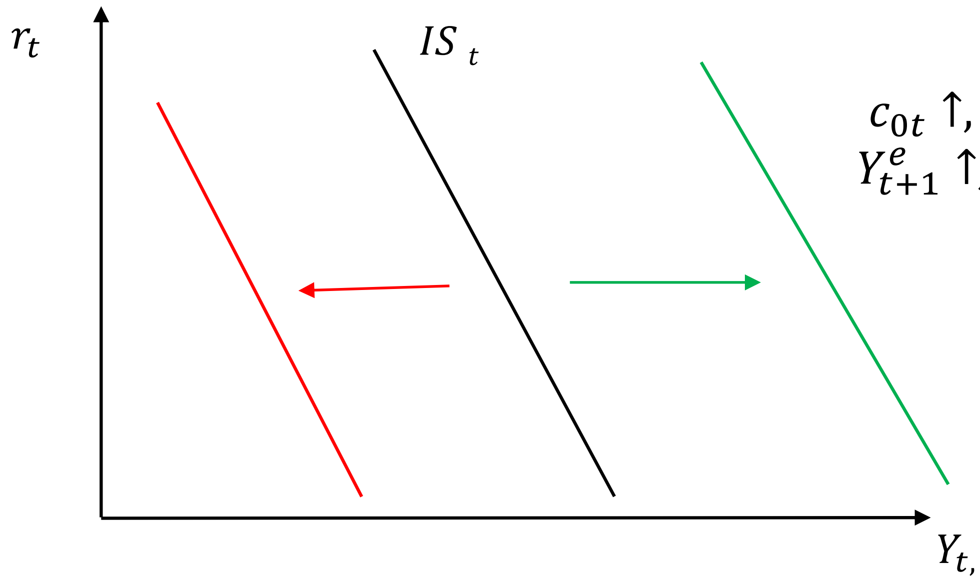


Negative slope but steeper

If $r \downarrow \Rightarrow A \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow$

But if we take expectations into account the increase is smaller since the future dynamics of the financial markets (r^e_{t+1}) and the of the goods markets (Y^e_{t+1}) also matter in determining current demand

Position of the current IS depends on both current and future exogenous variables



$c_{0t} \uparrow, T_t \downarrow, x \downarrow, \delta \downarrow \Rightarrow \; A_t \uparrow$
$Y^e_{t+1} \uparrow, T^e_{t+1} \downarrow, r^e_{t+1} \downarrow \Rightarrow A_t \uparrow$

The IS curve moves to the right if there is a positive increase in current demand dependent on

- Directly driven by changes in current exogenous variables

If $c_{0t} \uparrow, T_t \downarrow, x \downarrow, \delta \downarrow \Rightarrow \; A_t \uparrow \Rightarrow$ aggregate demand increases, in the goods market there is an excess demand, and in order to restore the equilibrium through the multiplier, production increases

- indirectly driven by changes in expectations

If $Y^e_{t+1} \uparrow, T^e_{t+1} \downarrow, r^e_{t+1} \downarrow \Rightarrow \; A_t \uparrow \Rightarrow \; Z_t \uparrow$ current aggregate demand $\Rightarrow$ thanks to the multiplier $Y_t \uparrow$

The IS curve shifts to the left if there is a contraction in current demand t : current period

The money market is not affected by expectations

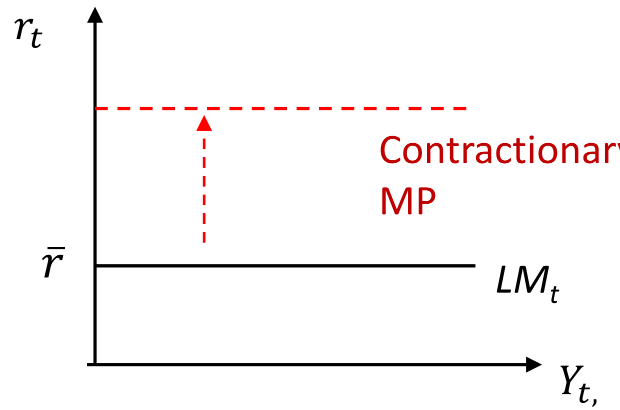
- Standard case: the CB sets the nominal interest rate $\bar{i}$ (short term interest rate i_{1t}) that keeps the money market in equilibrium changing the money supply

$$M^S = M^D$$

<u>Money supply:</u> depends on
- OMO
- Reserve ratio
- Fraction od money held as currency

<u>Money demand:</u> depends on
- Current level of transaction
- Current opportunity cost of money

Given expected inflation, the LM will be
$$r = \bar{r}$$



- Alternative case: the CB sets the money supply, and the interest rate i (short term interest rate i_{1t}) is the one that clears the money market

=> Upward sloping LM

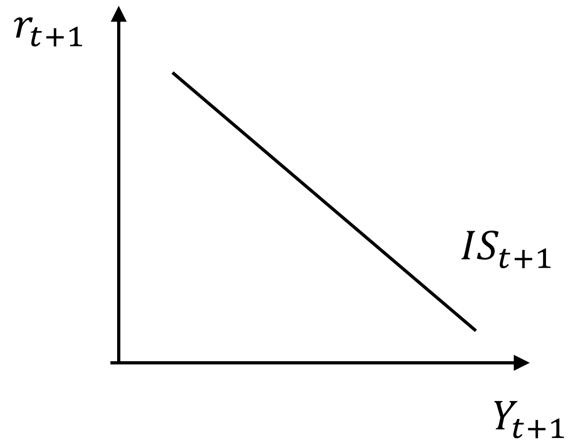
t+1 : future period

Being the last period only contemporaneous variables matter

- IS $_{t+1}$: Pairs of GPD and short term real interest rate allowing for the equilibrium in the goods market

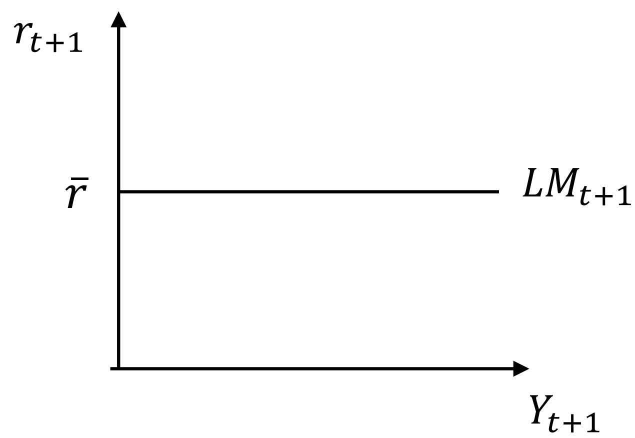
$$Y_{t+1} = \underbrace{C_{t+1} + I_{t+1}} + G_{t+1}$$

A_{t+1} = Future aggregate private expenditure: depends on future contemporaneous variables



$$Y_{t+1} = C\left(Y_{t+1,}, T_{t+1}\right) + I\left(Y_{t+1,}, r_{t+1}, x_{t+1}, \delta\right) + G_{t+1}$$

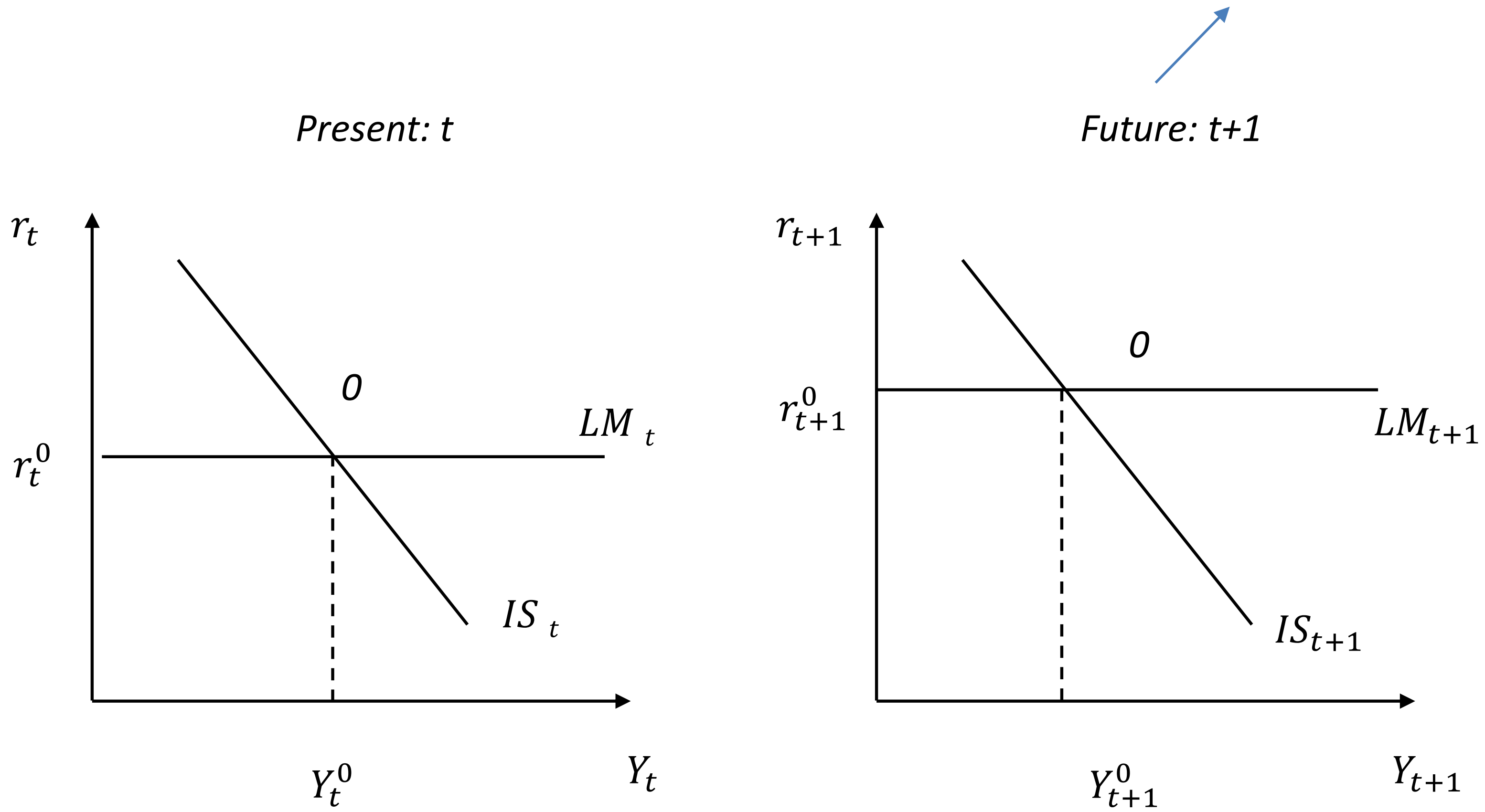
- LM $_{t+1}$: represents the future equilibrium for the money market (standard or alternative case)



The macro equilibrium today and tomorrow

All markets in equilibrium in the current and future period

It represents what agents expect at time t for the next period



THE IS – LM MODEL IN ACTION: HOW TO USE IT?

AIM: to show what happens to the macro equilibrium today, if we expect something for the future

1. Plot the initial equilibrium at t and t+1 and the yield curve

2. Begin always by studying the impact of the shock/policy at $t+1$ (future)

   How does the expected future equilibrium change?

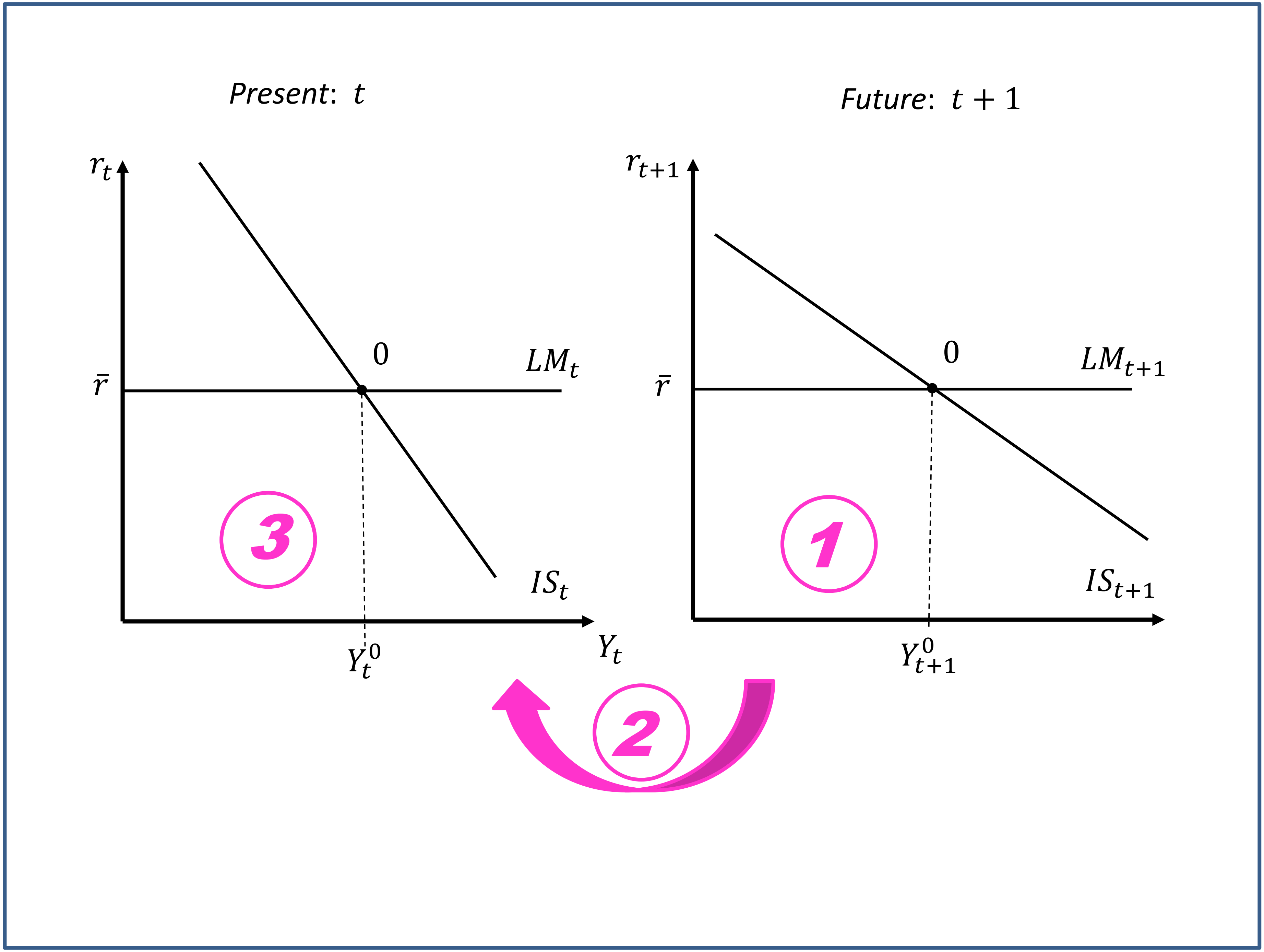
   $$\Delta Y^e_{t+1}\ ?\ \Delta r^e_{t+1}?$$

3. Study the impact today (t) of the policy/shock, incorporating changes in expectations

   How does the current equilibrium change?

   $$\Delta Y_t ?\ \Delta r_t$$

3. Analyse the impact on the yield curve



Present: t
Future: $t+1$
r_t
$\bar{r}$
0
LM_t
IS_t
Y_t^0
Y_t
3
r_{t+1}
$\bar{r}$
0
LM_{t+1}
IS_{t+1}
Y_{t+1}^0
1
2

Assumptions

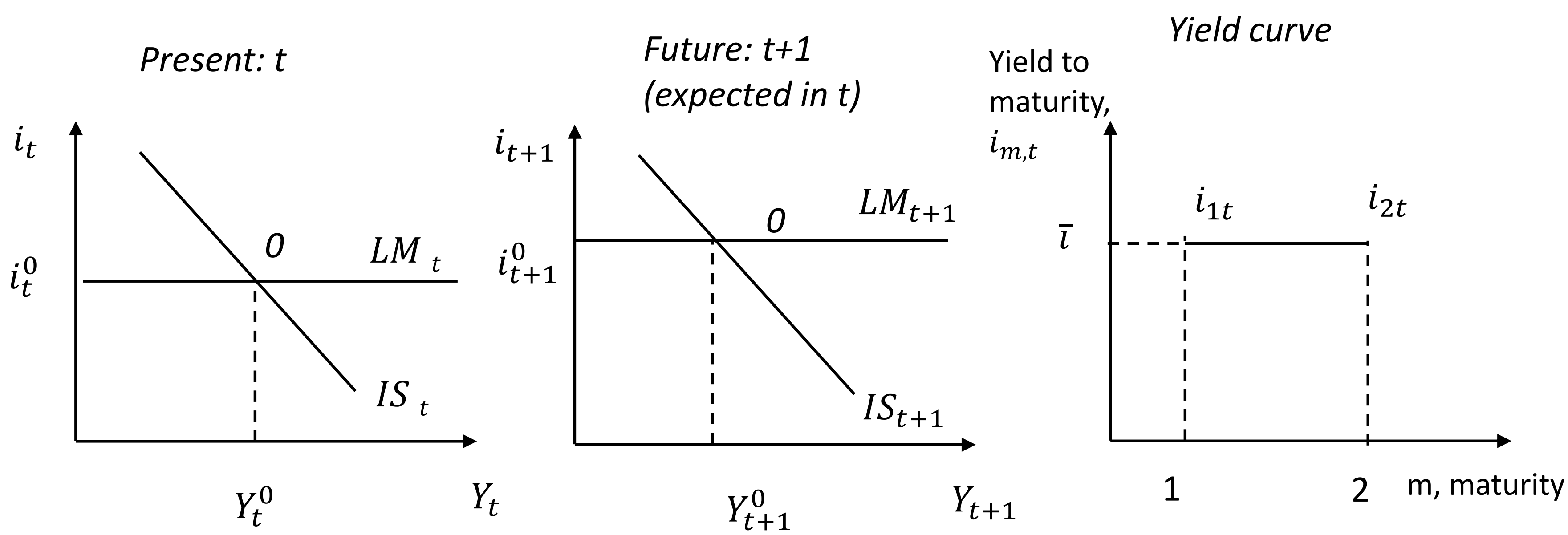
- Economy in initial equilibrium at t and t+1

- In both periods nor actual or expected inflation: $\pi_t = \pi^e_{t+1} = 0 \Rightarrow r = i$

- Agents risk neutral: no risk premium in the bond market

- CB has the same policy rate target in both periods $$\bar{\imath} = i_{1t} = i^e_{1t+1}$$

$$\Rightarrow i_{2t} = \frac{1}{2}(\bar{\imath} + \bar{\imath}) = \bar{\imath}$$



MONETARY POLICY AND EXPECTATIONS

The role of central banks is not only to adjust the interest rate, but above all to guide expectations.

Shaping expectations can be enough to produce the same effects as the actual implementation of a monetary policy operation.

Sometimes, just a few words can have a huge impact.

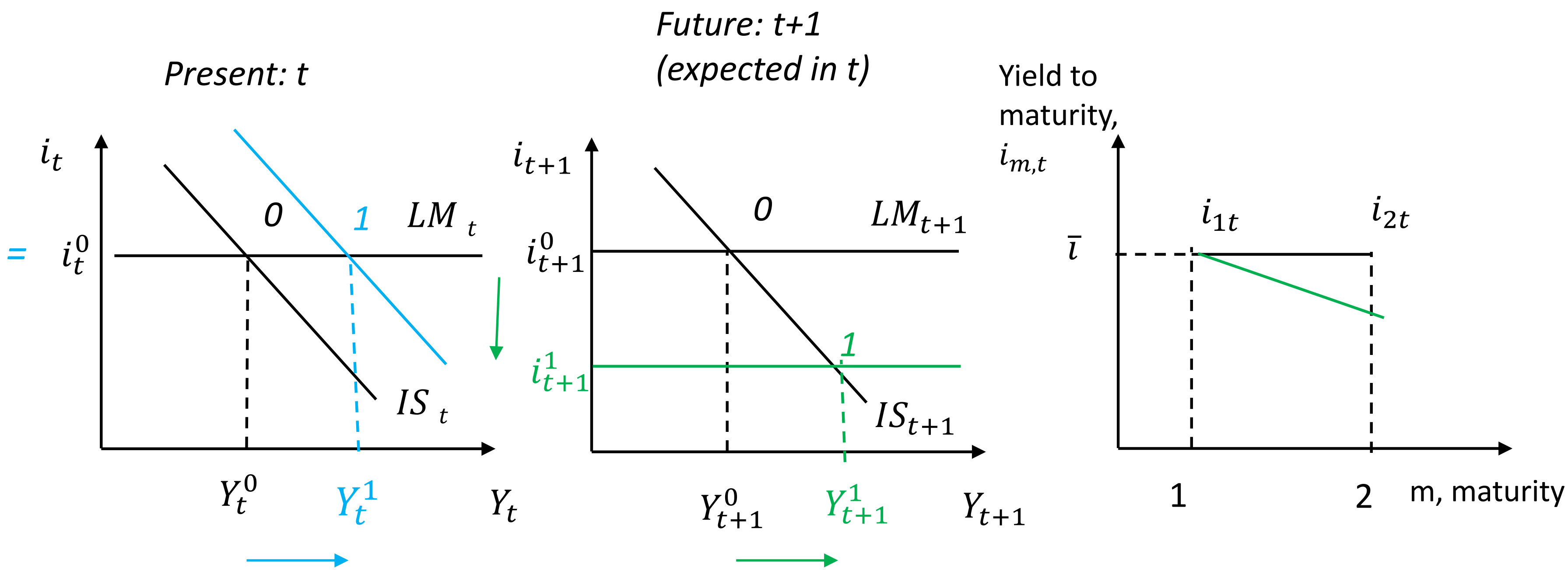
Mario Draghi's "Whatever it takes"

https://www.youtube.com/watch?v=tB2CM2ngpQg

CB announces its willingness to purchase bonds in the future, in order to address the recession.

0 = initial equilibrium

1 = final equilibrium



Indirect effect of the change in expectations at t+1

Direct effect of Expected expansionary MP implemented at t+1

Economic explanation

- Agents expect that in t+1, the CB will lower the nominal interest rate. Given the resulting lower real interest rate, investment will increase, stimulating demand and, through the multiplier effect, boosting output

LM_{t+1} down
Equilibrium 0 →1
$Y^e_{t+1} \uparrow, r^e_{t+1} \downarrow$

- In t, given their rational expectations, agents anticipate the effect expected for the future

• $Y^e_{t+1} \uparrow$
  • more disposable income expected for next $\Rightarrow LW_t \uparrow \Rightarrow C_t \uparrow$
  • $\Pi^e_{t+1} \uparrow, \mathrm{V}(\Pi^e) \uparrow$ investment more profitable $\Rightarrow I_t \uparrow$
  • Stocks more attractive as $€D^e_{t+1} \uparrow \Rightarrow €Q_t \uparrow \Rightarrow FW_t \uparrow$

$\Rightarrow$ aggregate private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

• $r^e_{t+1} \downarrow$
  • stocks more attractive $\Rightarrow €Q_t \uparrow \Rightarrow FW_t \uparrow$
  • firms more willing to invest $\Rightarrow \mathrm{V}(\Pi^e) \uparrow \Rightarrow I_t \uparrow$

$\Rightarrow$ aggregare private expenditure increases $\Rightarrow \mathrm{Z}_t \uparrow$ and through the multiplier $Y_t \uparrow$

Overall current output increases, the $M^D \uparrow$ and the CB to keep the interest rate constant at the chosen policy rate, $\uparrow M^S$

IS_t right
Equilibrium 0 →1
$Y_t \uparrow, i_t =$

Effect on the yield curve:

- Same starting point, as the short term interest rate does not change

- The slope becomes negative:

agents expect future short term interest rate lower than the current one

$\Rightarrow$ there is a negative spread btw the future expected short term interest rate and the current one

$$i_{t+1}^{1} < i_{t}^{0}$$

$\Rightarrow$ The long term interest rate will be lower that the initial one but still higher than the newly expected short-term rate

$$i_{2t} < \bar{\imath}$$
$$i_{2t} > i_{t+1}^{1}$$

The CB after having announced that it would implement an expansionary MP in the future, unexpectedly lowers the current interest rate.

Two Bloomberg analysts discuss the possible effects:

Analyst 1: *"GDP will definitely increase, and the yield curve will become downward sloping."*

Analyst 2: *"It's not certain that GDP will increase, and we can't draw any definite conclusions about the slope of the yield curve."*

So, who is right?

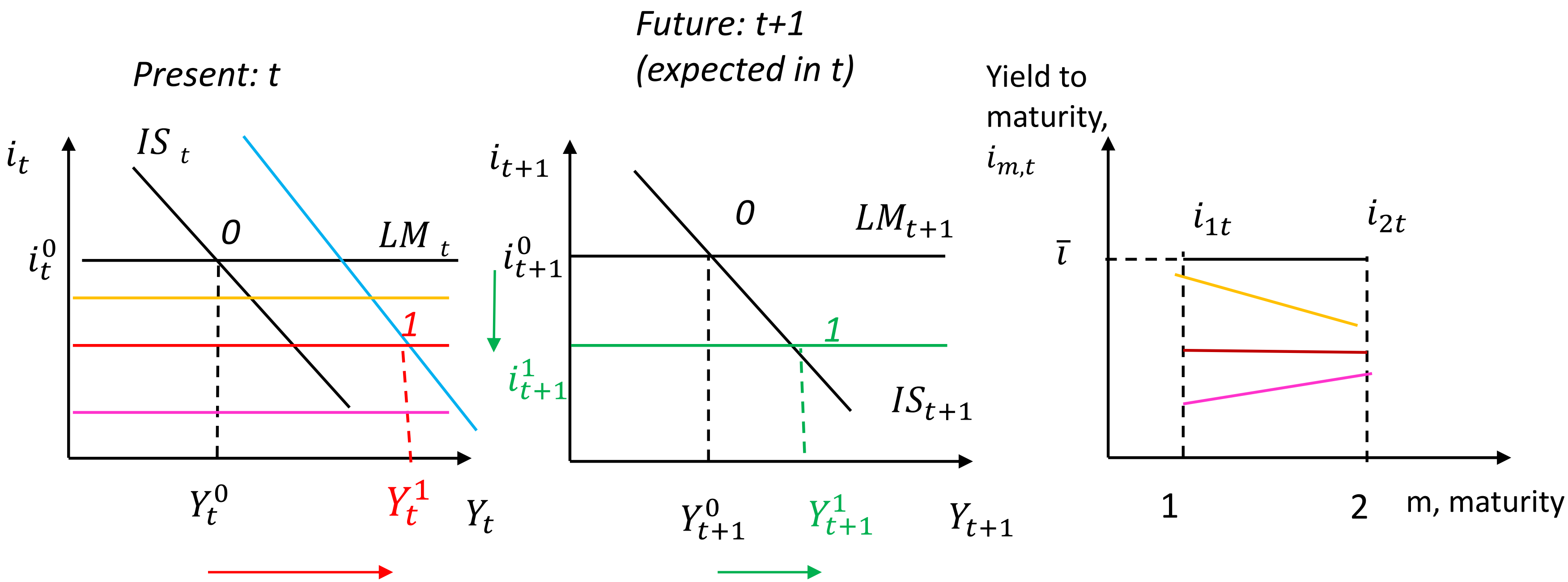
→ *Neither is completely right.*

It's likely that GDP will increase, but the effect on the yield curve is uncertain.

Expected expansionary MP
Current expansionary MP

0 = initial equilibrium

1 = final equilibrium



Indirect effect of the change in expectations at t+1

Direct effect of Expected expansionary MP implemented at t+1

The relative reduction in the current interest rate compared to the future rate influences the shape of the yield curve.

Effects

- If in t+1 the central bank lowers the nominal interest rate, by buying bonds, it will stimulate the real economy through investment, which becomes cheaper , Z↑ and through the multiplier effect Y↑.

LM_{t+1} down
Equilibrium 0 →1
$Y^e_{t+1}\uparrow, i^e_{t+1}\downarrow$

- In t, the equilibrium will change as a result of

1) Indirect effects of the changes in expected variables for the future that affect the IS

$Y^e_{t+1}\uparrow$

- more disposable income expected for next $\Rightarrow LW_t\uparrow\Rightarrow C_t\uparrow$
- $\Pi^e_{t+1}\uparrow, \mathrm{V}(\Pi^e)\uparrow$ investment more profitable $\Rightarrow I_t\uparrow$
- Stocks more attractive for investors as $€D^e_{t+1}\uparrow\Rightarrow €Q_t\uparrow\Rightarrow FW_t\uparrow$

$r^e_{t+1}\downarrow$

- stocks more attractive $\Rightarrow €Q_t\uparrow\Rightarrow FW_t\uparrow$
- firms more willing to invest $\Rightarrow \mathrm{V}(\Pi^e)\uparrow\Rightarrow I_t\uparrow$

IS_t right

$\Rightarrow$ ↑aggregare private expenditure increases $\Rightarrow \mathrm{Z}_t\uparrow$ and through the multiplier $Y_t\uparrow$

LM_tdown

2) Direct effect of current expansionary monetary policy: as the central bank reduces the nominal interest rate, the borrowing cost decreases, making investment less expensive ⇒ investment increases, stimulating demand and, in equilibrium, output.

Equilibrium at time t 0 →1
$Y_t\uparrow, i_t\downarrow$

Effects

- A permanently expansionary monetary policy has a **greater impact on output** compared to a policy implemented for only one period (either today or expected in the future), as it affects demand through two channels

- **Yield curve**
    - The starting point will definitely be lower, as the current monetary policy reduces the short-term interest rate.

    - The slope depends on the intensity of the current MP compared with the expected future one

    ⇔ it depends on the difference we expect btw i_t and i_{t+1}^e

    ⇔ If $i_t = i_{t+1}^e$ flat
    ⇔ If $i_t < i_{t+1}^e$ positive slope
    ⇔ If $i_t > i_{t+1}^e$ negative slope

    - For sure, the new long-term interest rate will be lower than the initial one, given that both the short-term interest rates at time t and time t+1 are lower, and so is their average.

FORWARD GUIDANCE OF THE CB

- What is it?
  Non conventional MP tool
  - A communication strategy used by central banks to influence market expectations about the future path of interest rates
  - The CB provides information about its future monetary policy intentions

- Why is it used?
  To lower long-term interest rates in order to stimulate I, Z and Y, when short-term rates are at or near the zero lower bound.

- When and Where?
  - Fed: Introduced in 2008, used extensively from 2011
  - ECB: Officially adopted on July 4, 2013

https://www.ecb.europa.eu/ecb-and-you/explainers/tell-me/html/what-is-forward_guidance.en.html

Suppose the markets are expecting a servere recession for the next year. Could the CB prevent the change in current output?

As we expect lower income in the next period, in current period

- C ↓ since our lifecyle wealth decreases
- I ↓ due to lower expected profits, making firms more cautious about investing

⇒ Aggregate demand falls

⇒ lowering, through the multiplier effect, output and income

To prevent the decline in current income the CB can use forward guidance on short term interest rates

⇒ The CB reassures markets about its future interventions to support the real economy by keeping short term interest rates low

⇒ If agents expect ↓ i^e_{t+1}, the PDV of expected profits ↑,

⇒ I_t ↑ => Z_t ↑

⇒ current output increases

Forward guidance helps to offset or eliminate the impact of adverse expectations

FISCAL POLICY AND EXPECTATIONS

Governments are not often forward-looking

- rarely act to reduce the deficit
- they mostly act to preserve political consensus

A contractionary fiscal policy:

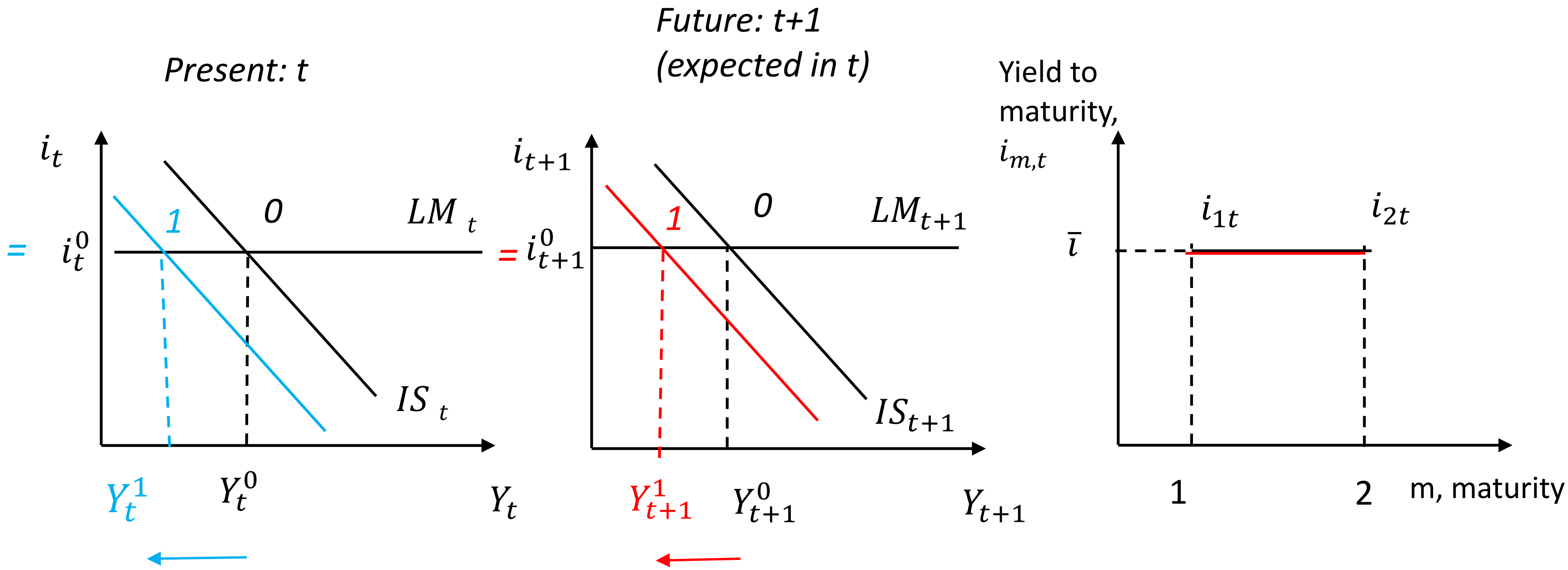
- In the short run, reduces output
- In the medium/long run, boosts savings, investment, and output

But using an intertemporal framework, outcomes could be different

Given the higher public debt, agents expect a future contractionary FP.

0 = initial equilibrium

1 = final equilibrium



Indirect effect of the change in expectations at t+1

Direct effect of contractionary FP to be implemented at t+1

Effects

- Agents expect for t+1 an increase in taxes or a reduction in public expenditure. The aggregate demand will be lower Z↓ and through the multiplier effect also output Y ↓. Given the lower money demand the CB will intervene by decreasing the money supply to keep the interest rate constant

IS_{t+1} to the left
Equilibrium 0 →1
$Y^e_{t+1}\downarrow,\quad i^e_{t+1} =$

- In t, the equilibrium will change as a result of

1) Indirect effects of the changes in expected variables for the future that affect the IS

$Y^e_{t+1}\downarrow$

- less disposable income expected for next $\Rightarrow LW_t \downarrow \Rightarrow C_t \downarrow$
- $\Pi^e_{t+1}\downarrow, \mathrm{V}(\Pi^e)\downarrow$investment less profitable $\Rightarrow I_t\downarrow$
- Stocks less attractive for investors as $€D^e_{t+1}\downarrow\Rightarrow €Q_t\downarrow\Rightarrow FW_t\downarrow$

=>↓aggregare private expenditure decreases $\Rightarrow \mathrm{Z}_t\downarrow$and through the multiplier $Y_t\downarrow$
=> In the momey market the money demand decreases and the CB to keep the interest rate constant reduces the money supply

IS_t left

Equilibrium 0 →1
$Y_t\downarrow, i_t =$

No changes in the yield curve, since both current and expected short-term interest rates remain unchanged
=> long-term rates are also unaffected

What would happen if the fiscal policy were implemented in the current period as well?

$\longrightarrow$ *Stronger real effect. Why ?*

$\longrightarrow$ *Same effect on the yield curve. Why ?*

What are the effects of a credible announcement of a future contractionary fiscal policy in an economy where the central bank controls the money supply?

$\longrightarrow$ *Ambiguous effects. Why ?*

SUMMING UP

- Expectations are crucial for understanding the effects of shocks and policies today
  - Expectations shape current economic outcomes.
  - Future policies influence today’s consumption and investment decisions
- Credibility of policy announcements is key
  - Credible announcements can be as powerful as actual actions.
- Forward guidance influences outcomes by shaping expectations, not through immediate action
- The yield curve reflects both present and future expectations.
- Takeaway: *It's not just what policymakers do, but what people expect they'll do*

Two economists, Elena and Marco, are discussing the consequences of expectations regarding a fiscal tightening planned for next year.

The government has announced that in period t+1, in order to avoid a public deficit, it will increase taxes and reduce public spending by an equivalent amount.

Elena argues: *“If the central bank wants to avoid the impact of the fiscal tightening on output both today (t) and tomorrow (t+1), then stock prices will rise and the yield curve will shift downward.”*

Marco, on the other hand, claims:
“If the central bank only responds to offset the change in output in period t+1, then stock prices might not change, but the yield curve will become downward sloping.”

Who is right?

No one

QUICK CHECKS

In the central bank of a common monetary area buys bonds, output increases _____ in countries where consumers ______ foresighted in their consumption decisions.

a) more, are
b) less, are not
c) more, are not
d) none of the previous answers

TRUE OR FALSE

Consider an economy with an initially flat yield curve. The government announces a future increase in public investment, while the central bank is expected to intervene to prevent changes in both current and future inflation. If agents are forward-looking and the announcement is credible, the yield curve is most likely to remain flat.

FALSE

We have contrasting effects but we do not know their relative magnitude : we are not able to predict the change in i_{2t}

QUICK CHECKS

The government announces an increase in taxes in the next period. Today, output will____________ if the central bank in the next period conducts monetary policy ___________ .

a) increase more, endogenously
b) increase less, exogenously
c) decrease more, endogenously
d) decrease less, exogenously

QUICK CHECK

The central bank announces an expansionary monetary policy in period t, and the policy is perceived as permanent. If the yield curve was flat before the shock, it will

a) become flatter and shift downward

b) become steeper and shift downward

c) remain flat and shift downward

d) none of the previous answers

TAKE AWAY AND KEY CONCEPTS

- Expectations
- Two periods IS-LM model and intertemporal framework
- Policy credibility
- Forward guidance
- Expected monetary policy
- Expected fiscal policy
- Permanent vs. temporary policies